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Subject	**:**	**Fundamentals of Telecommunication Systems**
Experiment Number :		**03**
Experiment Title	**:**	**Simulation of Signal Transmission through Wired Media**
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Experiment No. 03

Title:

Simulation of Signal Transmission through Wired Media.

Aim:

To study and simulate the characteristics of signal transmission through a lossy wired medium (Coaxial Cable) using LTspice and observe the effects of attenuation and propagation delay.

Software Used:

LTspice 26.0.1 (High-performance SPICE simulation software).

Theory:

Wired Media Characteristics: In telecommunication, physical cables like Coaxial or Twisted Pair are not ideal conductors. They possess electrical properties distributed along their length: Resistance (R), Inductance (L), Capacitance (C), and Conductance (G).

The LTRA Model: The Lossy Transmission line (LTRA) model in LTspice simulates real-world behaviour.

Attenuation: The signal strength decreases over distance due to Resistance (R).

Distortion: The signal shape changes (rounding of edges) due to Capacitance (C) and Inductance (L).

Propagation Delay: It takes a finite amount of time for the signal to travel from the source to the destination.

Simulation Procedure

Step 1: Component Selection and Placement

Open LTspice and create a New Schematic.

- *Voltage Source:* Select voltage, and place it. Right-click it, select Advanced, then PULSE. Set: Vinitial=0, Von=5, Tdelay=0, Trise=1n, Tfall=1n, Ton=0.5u, Tperiod=1u.
- *Transmission Line:* Press ‘P’ search for the component ‘ltline’. Select it and place it on the schematic. It will appear with the designator O1 and LTRA.

- *Load Resistor:* Press ‘R’, place a resistor at the output. Right click on the resistor symbol and set resistance value to 50 ohms (to match the characteristic impedance of co-axial cable).

Step 2: Defining the Cable Model

- Right-click the ltline (O1) component. In the Value field, replace LTRA by typing ‘MyCo- axCable’.
- Click the ‘.t’ button (SPICE Directive) on the top toolbar.
- In the dialog box, paste the following command exactly: .model MyCo-axCable LTRA(R=0.1 L=250n C=100p G=0 len=100)
- Place this text anywhere on your schematic area. This defines MyCo-axCable as a 100-meter lossy line.

Step 3: Wiring and Grounding

- Press ‘G’ to place grounds. You must ground the bottom terminal of the Voltage Source, the two bottom terminals of the ltline (O1), and the Load Resistor.
- Press ‘W’ to wire the components. Connect all points properly.
- Use ‘t’ to label the input wire as IN and the output wire as OUT.

Step 4: Setting up the Simulation

- Click configure analysis (.ac) from the toolbar or type ‘A’ to configure transient analysis.
- In the Transient tab, set the Stop Time to 5u.
- Place the .tran 5u command anywhere on the schematic.
- Click the Run (Running Man) icon.

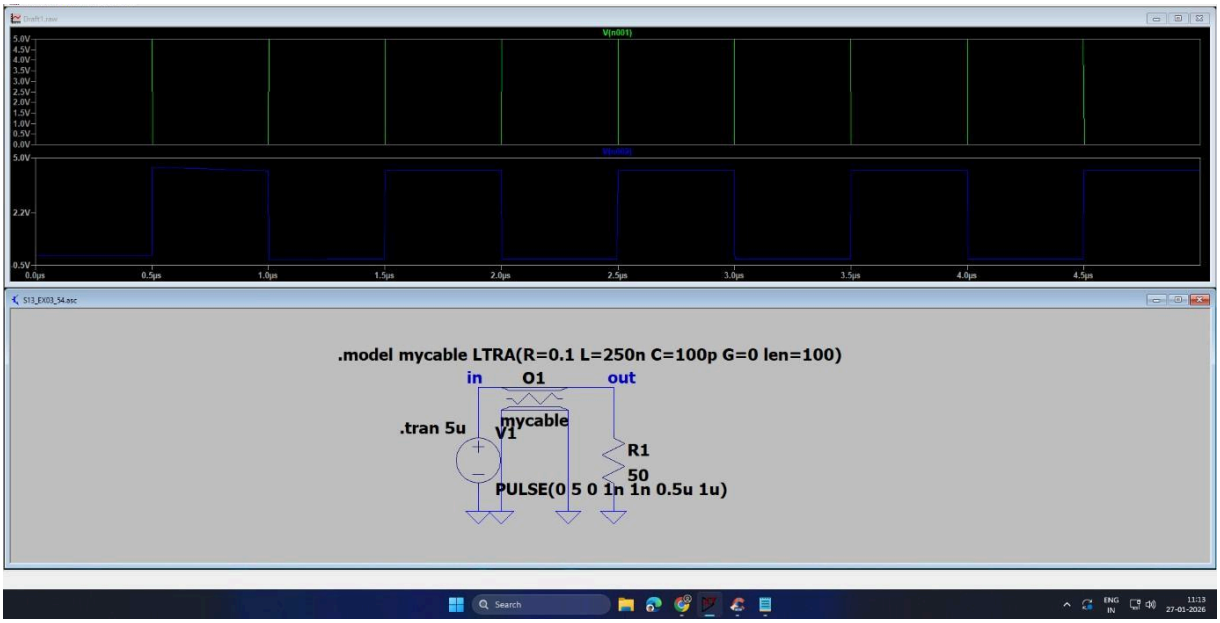
Observation and Analysis

Students must perform the simulation for the following three cases by right-clicking the .model directive and changing the len value each time.

Case 1: Cable Length = 100 meters

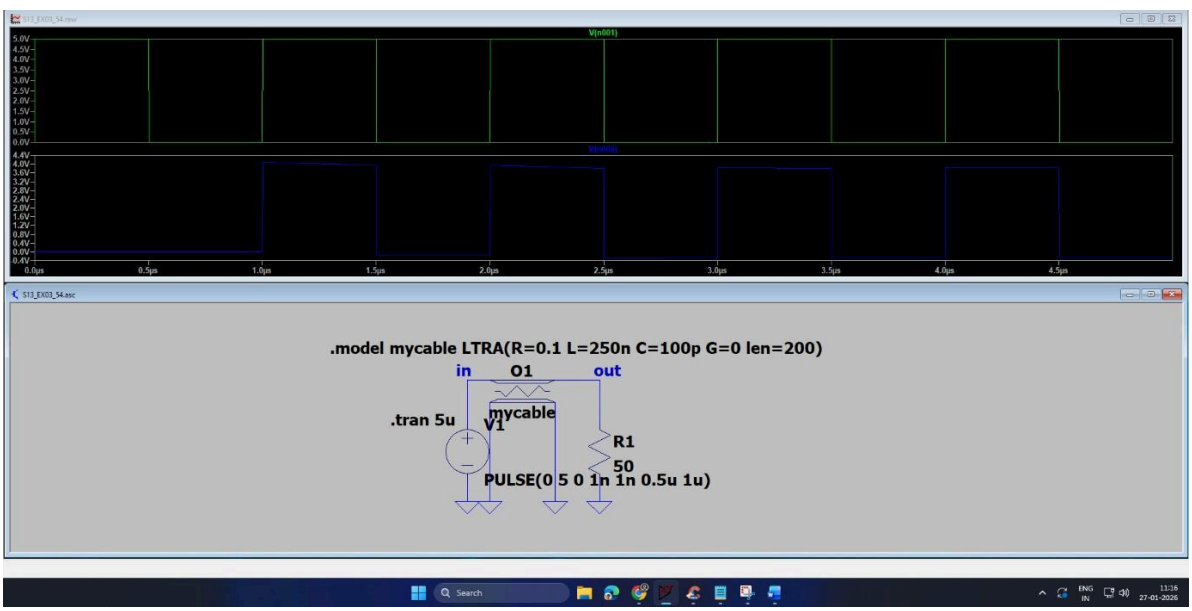
- Task: Run simulation with len=100.

- Screenshot:
- Data: Record the peak output voltage and the time delay.



Case 2: Cable Length = 200 meters

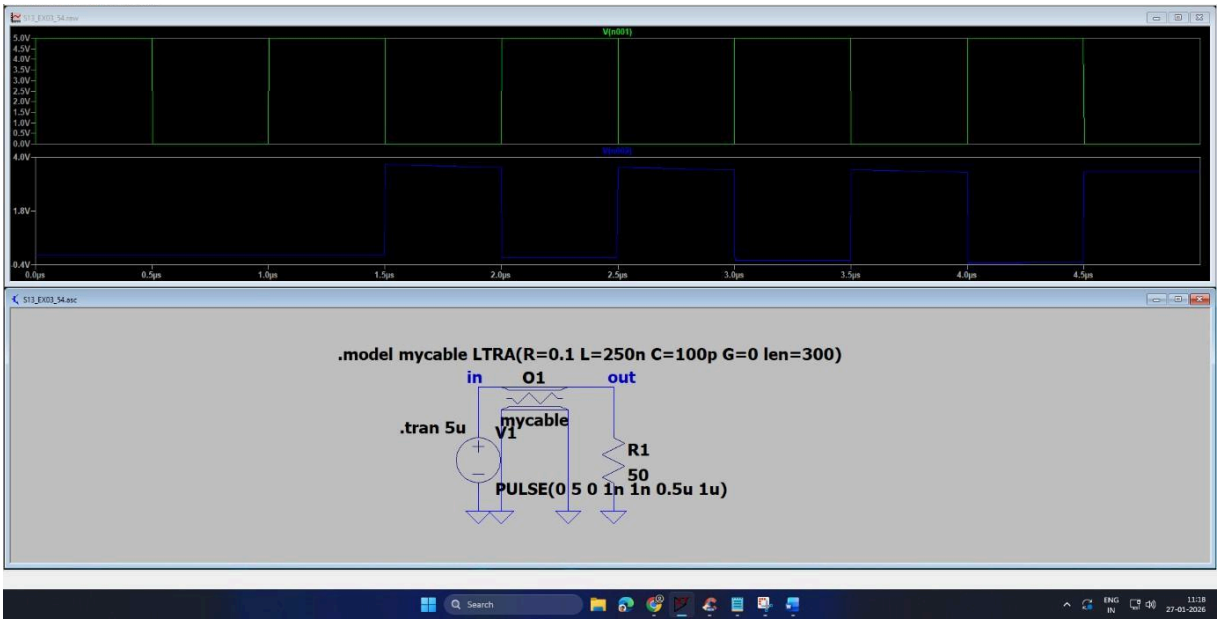
- Task: Update directive to len=200 and re-run. Change the SPICE Directive to define cable length of 200 meters
  .model MyCo-axCable LTRA(R=0.1 L=250n C=100p G=0 ***len=200****)*
- Screenshot:



- Data: Record the peak output voltage and observe the increased delay.

Case 3: Cable Length = 300 meters

- Task: Update directive to len=300 and re-run. Change the SPICE Directive to define cable length of 300 meters
  .model MyCo-axCable LTRA(R=0.1 L=250n C=100p G=0 ***len=300****)*
- Screenshot:
- Data: Record the peak output voltage and observe the increased delay.



Observation Table

Case	Cable Length	Input Voltage	Output Voltage	Propagation Delay
1	100 m	5V	-4.8V	~0.5 µs
2	200 m	5V	-4.4V	~1.0 µs
3	300 m	5V	-4.0V	~1.5 µs

Calculations:

Calculate the percentage attenuation using the formula:

$$Attenuation\ \% = \frac{V_{in} - V_{out}}{V_{in}} x\ 100$$

Conclusion

The simulation demonstrates that wired media is not ideal. As the cable length increases, attenuation and propagation delay become more prominent. For the cable length of 100 m, 200 m and 300 m the attenuation observed are 4%, 12%, and 20% respectively whereas propagation delay observed was 0.5 µs, 1.0 µs and 1.5 µs respectively.